

Experiment 16:

HIGH LEVEL PROGRAMMING EXTENSIONS - PROCEDURES

AIM:

To implement procedures using program in MySQL.

PROCEDURES:

A procedure is a subprogram that performs a specific action.

Creating a procedure:

We use the CREATE PROCEDURE statement to create a new stored procedure. We specify the name of stored procedure after the CREATE PROCEDURE statement. The DELIMITER command is used to change the standard delimiter of MySQL commands (i.e. ;). As the statements within the routines (functions, stored procedures or triggers) end with a semi-colon (;), to treat them as a compound statement we use DELIMITER.

Calling stored procedures(Executing a procedure)

In order to call a stored procedure, you use the following SQL command:

CALL stored_procedure_name();

Program 1:

```
mysql> CREATE TABLE student_info (
->     id INT,
->     name VARCHAR(50),
->     department VARCHAR(30)
-> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO student_info VALUES
-> (1, 'Arun', 'CSE'),
-> (2, 'Bala', 'ECE'),
-> (3, 'Priya', 'IT');
Query OK, 3 rows affected (0.05 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> CALL GetStudents();
+----+-----+-----+
| id | name  | department |
+----+-----+-----+
| 1  | Arun  | CSE       |
| 2  | Bala  | ECE       |
| 3  | Priya | IT        |
+----+-----+-----+
3 rows in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)
```

Program 2:

```
mysql> SELECT * FROM customers;
+-----+-----+-----+
| customerNumber | customerName | creditLimit |
+-----+-----+-----+
|          101 | Arun         | 60000.00   |
|          102 | Bala         | 30000.00   |
|          103 | Priya        | 5000.00    |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE GetCustomerLevel(
-> IN p_customernumber INT,
-> OUT p_customerlevel VARCHAR(20)
-> )
-> BEGIN
-> DECLARE v_creditlimit DECIMAL(10,2);
->
-> SELECT creditLimit
-> INTO v_creditlimit
-> FROM customers
-> WHERE customerNumber = p_customernumber;
->
-> IF v_creditlimit > 50000 THEN
-> SET p_customerlevel = 'PLATINUM';
-> ELSEIF v_creditlimit >= 10000 THEN
-> SET p_customerlevel = 'GOLD';
-> ELSE
-> SET p_customerlevel = 'SILVER';
-> END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> SET @level = '';
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CALL GetCustomerLevel(101, @level);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT @level AS Customer_Level;
+-----+
| Customer_Level |
+-----+
| PLATINUM       |
+-----+
1 row in set (0.00 sec)

mysql> SET @level = '';
Query OK, 0 rows affected (0.00 sec)

mysql> CALL GetCustomerLevel(102, @level);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT @level AS Customer_Level;
+-----+
| Customer_Level |
+-----+
| GOLD           |
+-----+
1 row in set (0.00 sec)

mysql>
mysql> SET @level = '';
Query OK, 0 rows affected (0.00 sec)

mysql> CALL GetCustomerLevel(103, @level);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT @level AS Customer_Level;
+-----+
| Customer_Level |
+-----+
| SILVER         |
+-----+
```